Deploying Mojaloop 17.x.0 and upgrades

MojaCom31, Online meetings, 15th July 2026

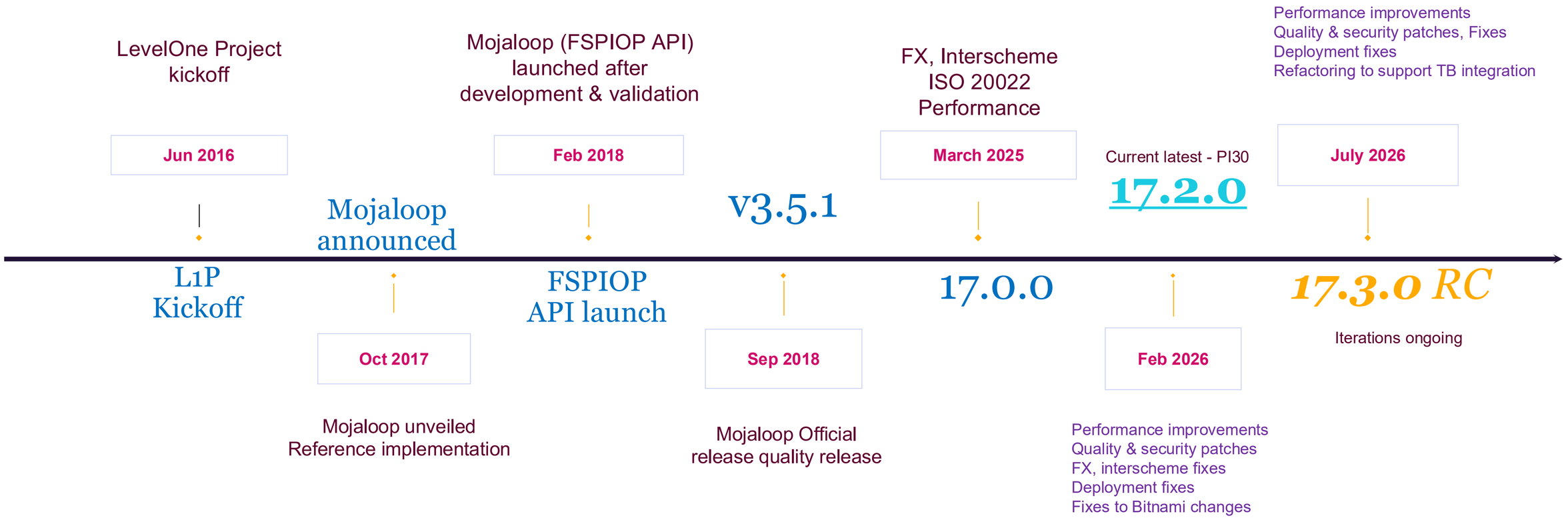
Sam Kummary, Mojaloop Foundation

Agenda

1. Mojaloop Releases Overview
    a. v17.2.0 Features, updates
    b. V17.3.0 Release Candidate (RC)
    c. Platform quality & security
    d. Upgrades and PQS discussion (Round table - Q&A)
2. Whats next?

Mojaloop Releases

PI-31



** Version numbers guided by SemVer*

Mojaloop Platform

1. Mojaloop Core

   Transactional hub components and settlement i.e. API layer, central ledger (including bulk), settlement, ALS, quoting, trx req, Admin

2. Mojaloop Connector(s)

   Best practice Mojaloop connectors. I.e. Third-party services, sdk adapter/ml-connectors

3. Supporting services

   Interop: Connect with settlement banks, Onboarding FSPs, integrations (connection management)

4. Mojaloop Toolkit (TTK), ITK instances, Simulators, ALS support, CEP, Events processing

   Oracles - MSISDN or Merchant Registries

5. Finance portal (RBAC), Reporting

   Hub operator UI & API experience layer that is Implemented by business operations framework.

6. Monitoring support (Operational support)

   Operational support & tracing (e.g. EFK / Prometheus, Grafana, Loki, etc) [IaC: Grafana, Prometheus & Loki]

7. API Gateway

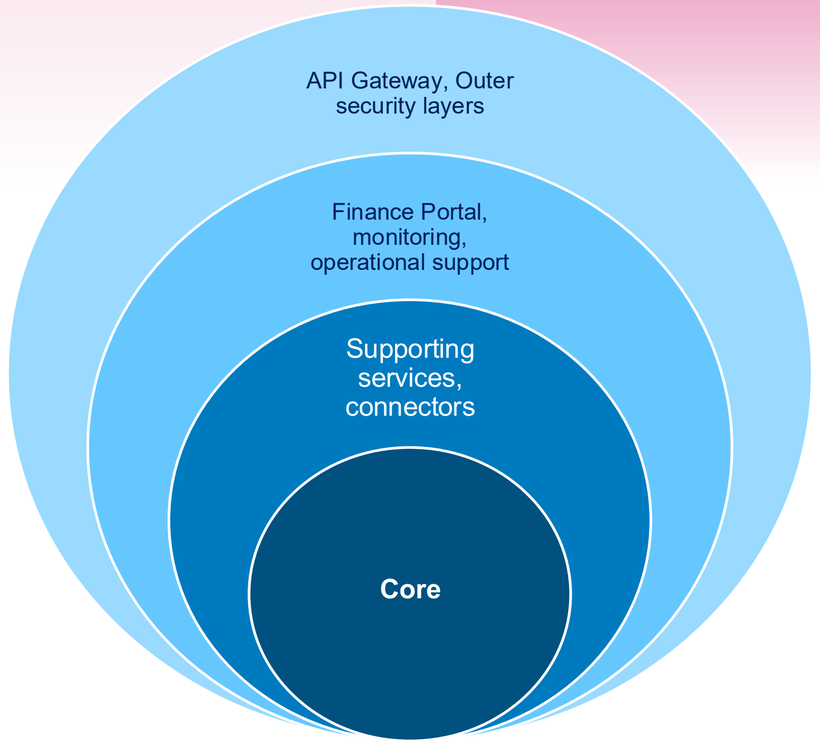
   IP filtering, Ingress/Egress, Firewalls, Security support (JWS, mTLS) [IaC provides this via Istio (older versions used WSO2)]

   Adds in the rules and enforces the Mojaloop Security design

8. Infrastructure

   Kubernetes, Infrastructure as Code, Automated deployments (IaC)

   Deployment artifacts - Helm Charts, Docker Containers, npm packages



Mojaloop IaC bundle

Bundled Mojaloop IaC Release (Mojaloop IaC Release 5.x.x / 6.x.x)

1. Gateway & Load Balancers
2. RBAC via Istio and Ory Stack
3. Mojaloop Security (mTLS, JWS, ILP)
4. Finance Portal (Business operations) – support for FX & inter-scheme
5. Mojaloop Connection Manager (MCM)
6. Reporting
7. Control Plane for CI/CD (GitLab) – CC v2
8. Technical Operational Dashboards (Monitoring)
9. Participant Onboarding Support (Connection Manager)
10. Optional Payment Manager deployment (for DFSPs)

Mojaloop v17.2.0

Mojaloop Release: Core

1. Base use-cases
    1. P2P, R2P, Bulk
    2. FX, Inter-scheme
    3. ISO 20022 messaging
    4. *Performance, quality & security improvements*
2. Settlement: Deferred-net, Continuous Gross, FX support
3. Admin: Participant Lifecycle API
4. Third-party payments v1 (PISP v2 development in progress)

Mojaloop Switch Functionality – FSPIOP Use Cases (PI-29)

Mojaloop FSPIOP API v1.1, v2.0 Use cases

Payer-Initiated Transaction

[●] P2P Transfers
[●] Customer-Initiated Merchant Payment
[●] Customer-Initiated Cash-out - Receive Amount
[●] Customer-Initiated Cash-out - Send Amount
[●] ATM-Initiated Cash-out
[●] Refund
[●] ***Foreign Exchange and Inter-scheme***

Bulk Transactions

[●] Bulk Payments

Payee-Initiated Transaction

[●] Merchant-Initiated Merchant Payment
[●] Agent-Initiated Cash-out
[●] Agent-Initiated Cash-In – Send Amount
[●] Agent-Initiated Cash-In – Receive Amount

Payee-Initiated Transaction using OTP

[●] Merchant-Initiated Merchant Payment Authorized on POS
[●] Agent-Initiated Cash-out Authorized on POS

Key
[●] Supported with functional tests
[●] Supported
[●] Proof of Concept
[●] Not implemented
[o] Out of Scope

Switch Functionality: Mojaloop End-points (PI-29)

FSPIOP API v1.1, v2.0 Specification

Transfers*
[●] POST - Prepare
[●] PUT - Response
[●] PUT – Error
[●] PATCH – Notification
[●] GET - Query

Parties
[●] GET - Request
[●] PUT - Response
[●] PUT - Error

Quotes
[●] POST - Request
[●] PUT - Response
[●] PUT - Error
[●] GET - Query

Participants
[●] POST - Create
[●] PUT - Response
[●] POST - Bulk Create
[●] PUT - Error
[○] DEL - Delete

Transactions
[○] PUT - Response
[○] GET - Query

TransactionRequests
[●] POST - Request
[●] PUT - Response
[●] PUT - Error
[●] GET - Query

Authorizations
[●] GET - Request
[●] PUT - Response
[●] PUT - Error

BulkTransfers*
[●] POST - Request
[●] PUT - Response
[●] PUT - Error
[●] GET - Query

BulkQuotes
[●] POST - Request
[●] PUT - Response
[●] PUT - Error
[●] GET - Query

fxQuotes*
[●] POST - Create
[●] PUT - Response
[●] GET - Query
[●] PUT - Error

fxTransfers*
[●] POST - Request
[●] PUT - Response
[●] PUT - Error
[●] GET - Query

*Enhancements in PI-26,27

Key
[●] Supported with functional tests
[●] Supported
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[○] Out of Scope

Switch Functionality – Administration (PI-29)

Admin API v1.0 Specification

Participants

[●] Manage Participants
- [●] Create Initial Values
- [●] Query
- [●] Update

[●] Manage positions, limits
- [●] Create Initial Value
- [●] Query
- [●] Update

[●] Manage Callback URLs
- [●] Create Initial Value
- [●] Query
- [●] Update

Settlement Models

[●] Settlement models
- [●] Configure
- [●] Query

Participant Accounts

[●] Manage accounts
- [●] Create
- [●] Query
- [●] Update
- [●] Funds In/Out

Transactions

[●] Transactions
- [●] Query

* Enhancements in PI-26,27

ALS Oracle API v1.0 Specification

Oracles (ALS)

[●] Manage Oracles
- [●] Create
- [●] Query
- [●] Update
- [●] Delete

Services

[○] Services FXP*
- [○] Get (Query)
- [○] PUT

Key
[●] Supported with functional tests
[●] Supported
[●] Proof of Concept
[●] Not implemented
[○] Out of Scope

Switch Functionality – Settlement (PI-29)

Settlement API v2.0 Specification

Settlements v**1.0** [Deferred, Multilateral Net]

[●] Open, close Settlement Windows
[●] Query Settlement Windows
[●] Query Settlement Report
[●] Create/Trigger Settlement with Windows
[●] Process successful Settlement Acknowledgements
[●] Reconcile Positions based on successful Settlements
[●] Process failed Settlement Acknowledgements

Settlements v**2**

[●] Settlement models
[●] Designs
[●] Settlement by Currency
[●] Interchange Fees
[●] Continuous Gross Settlement

Key
[ ] Supported with functional tests
[●] Supported
[●] Proof of Concept
[●] Not implemented
[○] Out of Scope

Switch – Third-party (3p) API v1.0 (PI-29)

Third-party API v1.0 Specification

Linking

[●] Pre-Linking
[●] Discovery
[●] Request consent
[●] Authentication
[●] Grant consent
[●] Credential registration

Un-Linking

[●] Un-linking

Third-party (3p) Transfers

[●] Discovery
[●] Agreement
  [●] 3p transaction request
  [●] 3p authorization
  [●] Signed authorization
  [●] Validate authorization
[●] 3p Transfer
[●] Query trxreq status

PISP *v2.0* (in progress)

[●] Discovery
[●] Agreement
  [●] 3p transaction request
  [●] 3p authorization
  [●] Signed authorization
  [●] Validate authorization
[●] 3p Transfer

Key
[●] Supported with functional tests
[●] Supported
[●] Implementation in progress
[●] Not implemented
[○] Out of Scope

WHY v17.2.0 or v17.3.0 RC?

1. Improved Performance and enhancements to existing features
2. Enhanced Security (Patching, package upgrades)
3. Bug Fixes and Stability (FX, inter-scheme, lookup, logging)
4. Fixes to helm registry issues (Bitnami changes)
5. Future-Proofing (Prepare for future releases - Improvements to logging, Tigerbeetle integration, more performance improvements)

Workstream progress can be tracked on Mojaloop project repo

Mojaloop v17.3.0 RC

Mojaloop Release: Core

1. Performance, quality & security improvements
2. Bug fixes
3. Deployment improvements
4. Maintenance improvements & refactoring to support future releases

Mojaloop v17.3.0 RC

Mojaloop Release: Platform quality & security

1. Hardening Mojaloop’s CI/CD Supply Chain
    1. mojaloop/project #4479 · PI30
    2. The audit team: @elnyry-sam-k · @kirgene · @shashi165 · @gibaros
2. CircleCI orb review & hardening: Checksum-verified downloads, scoped apt keys (in review), pinning-chain maintenance, SPDX gate semantics
3. Tooling + release paths: SBOM-first license gate rebuildl npm & Docker release-path analysis; trusted-publishers proposal
4. ci-config deprecation and rollout
5. CI runs moved from unpinned clone (repo: ci-config) to the mojaloop/build orb
6. License gate: @mojaloop/license-scanner-tool rebuilt to be SBOM-first (@shashi165 (#4481))

Upgrading Mojaloop Q&A



AbliPay, Infitx, WiredIn

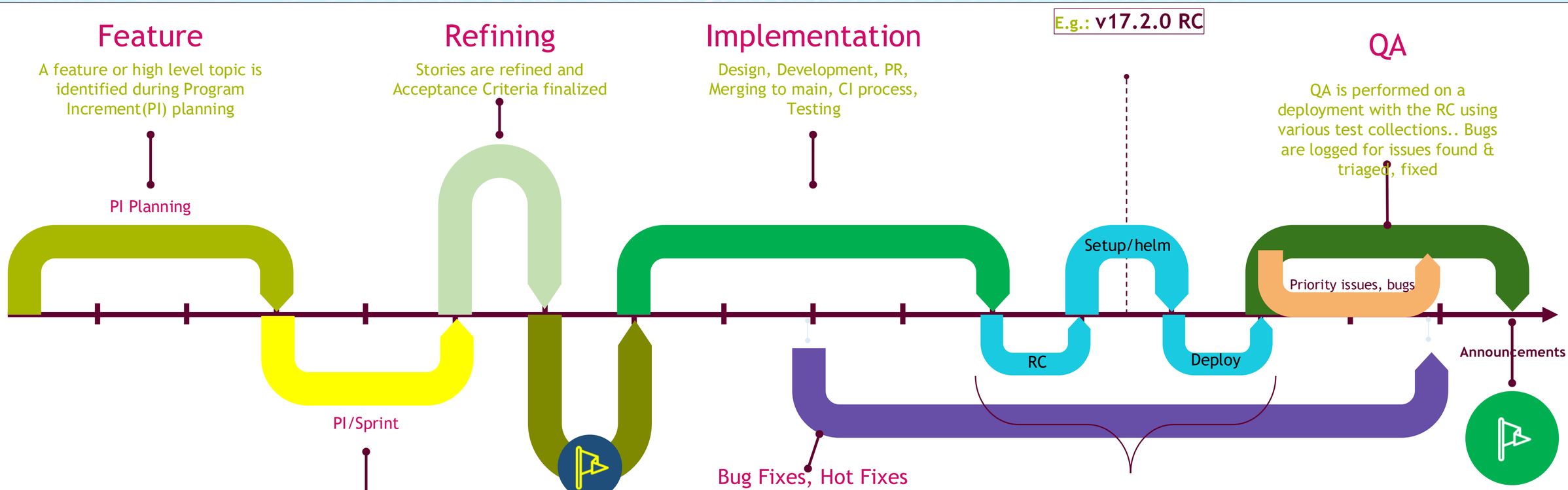
1. Which Mojaloop versions have you worked with, whats your favorite version? Any specifics on Mojaloop v17.2.0
2. How do you handle Continuous Integration, Continuous Deployment setup for your implementations / projects
3. Please share your experiences with deploying and also upgrading Mojaloop where applicable.

Mojaloop quality & security Q&A

AbliPay, Infitx, WiredIn

1. In your implementations and deployments, what does platform quality & security mean at a high level
2. How comfortable are you with Mojaloop platform’s quality & security
3. Any advice for current and future implementers (SIs and Adopters)

Mojaloop Release Process - overview



* SemVer x.y.z. Moved to pure Semver starting PI-11 (11.x.x)

Note: This depicts a single iteration that may happen multiple times

ML Releases: PI-24/27 update for reference

v15.1.0 (PI-21)	v15.2.0 (PI-22/23)	v16.0.0 (PI-23)	**v17.0.0** (PI-26)
1. Test against upgraded QA/Dev with K8s v1.26/1.27 2. Refactored QA/Dev env ease of maintenance with EKS 3. Enhanced SDK support for Request-to-Pay (R2P) 4. Follow-up bug-fixes 5. IaC v4.1.0 with support for Mojaloop v15.1.0	1. Upgraded QA/Dev env & Tested against K8s v1.28 2. General maintenance 3. Node upgrade (v18 LTS) 4. Performance setup, profiling and document bottlenecks (perf workstream) 5. Performance Stream improvements (enhancements, fixes, instrumentation, & contribs) 6. Automated, CI tests on ML releases with Helm snapshot support	1. General maintenance 2. Node upgrade for supporting services not updated in v15.2.0 3. Performance enhancements done by perf-workstream included 4. Automated CI steps on Mojaloop releases 5. IaC changes - new and refactored version	1. FX, Inter-scheme support 2. Support for ISO 20022 messaging 3. Merchant payments features 4. PQS updates - security patches and upgrades 5. Performance enhancements 6. General maintenance

Release timeline PI-30: MLF Product & work-streams

V17.2.0
PI 30

1. Performance improvements
2. Fixes to Bitnami changes
3. PQS - security patches & vulnerability management
4. Deployment improvements - Helm and IaC
5. Merchant services updates
6. General maintenance, bug fixes (FX, inter-scheme - improve discovery, logging, error-codes)

V17.3.0 RC
PI 31

1. Performance improvements (round#2)
2. Fixes to Bitnami changes
3. PQS - security patches & vulnerability management
4. Deployment improvements - Helm and IaC
5. General maintenance, bug fixes (FX, inter-scheme - improve discovery, logging, error-codes)

Future
v17.x.y / v18.0.0

1. Deployment improvements
2. Logging enhancements
3. Settlements improvements
4. TigerBeetle integration into Mojaloop Core
5. PISP Bulk API support (#107)
6. PISP v2 support
7. Enhance tracing components to use latest open-tracing standards
8. Automated CI testing on Helm PRs & Releases - completed
9. Migrate Charts (i.e. BoF) to Helm
10. Patch notification support for failures

Mojaloop Core & Releases - Ongoing activities

1. Iterating over Mojaloop v17.3.0 RC and preparing for TB integration
2. Performance whitepaper workstream support, performance improvements
3. Standardize helm charts to use standard charts for backend services (Kafka, mysql, mongo, redis)
4. MCM and Identity management enhancements
5. Mojaloop deployment enhancements, documentation updates, related testing and general maintenance (bug fixes, upgrades, patches)

Workstream progress can be tracked on Mojaloop project repo

Mojaloop PQS: In addition to CI steps

1. GitHub tools – CodeQL, Dependabots, Synk, Co-pilot secret scans
2. Peer reviews of contributions
3. Security audits every PI
4. Review security, quality processes during releases
5. Sonarcloud scanning
6. Test coverage standards
7. End-to-end test suites
8. Deployment validation
9. Release checklist

Thank you!

Resources:

1. Mojaloop docs: https://docs.mojaloop.io
2. Mojaloop training program: https://learn.mojaloop.io/
3. Mojaloop core & releases slack channel (pi29): https://mojaloop.slack.com/archives/C06VB2FHUER
4. Mojaloop platform quality & security slack channel (pi29): https://mojaloop.slack.com/archives/C06UW0E2KBN
5. Appendix following this

How we work

1. Messaging: Short-term - Slack; Community central - announcements
2. GitHub: Issues for longer form discussion; Issue tracking, Project
3. Documentation
    a. Markdown on GitHub for specifications and documents with long life.
    b. Sequence diagrams: Plant UML, DrawIO, etc…
4. Our Git process (forking/branching WF), the CLA
5. Quality gateways
6. Apache 2.0 License for code
7. API specifications: docs.mojaloop.io/api , Change requests, Solution Proposals
8. Scaled Agile Framework (SAFe)

Core & releases Mojaloop slack channel: https://mojaloop.slack.com/archives/C06VB2FHUER

Mojaloop

PHASES 1-2

2016	2016	2017	2018	2018	2018
JUNE	OCTOBER	OCTOBER	FEBRUARY	APRIL	SEPTEMBER
LevelOne Development project kickoff	API Specification project kick-off (PDP)	Mojaloop (branding) and OSS launch at SIBOS	Mojaloop API (FSPIOP) specification launch	First Mojaloop community event in Johannesburg	First Mojaloop official release v3.5.1 (pre-releases so far)

Mojaloop

PHASES 3-6

5.2.0

Sample release from February 2019 https://github.com/mojaloop/helm/releases/tag/v5.2.0

2019 Feb

10.1.0

Settlements enhancements

2020 May

10.4.0

FSPIOP v1.1 support for Mojaloop platform: https://github.com/mojaloop/helm/releases/tag/v10.4.0

2020 July

11.0.0

Semantic versioning standard adopted: https://github.com/mojaloop/helm/releases/tag/v11.0.0

2021 Jan

12.0.0

CGS, Interchange Fee supported: https://github.com/mojaloop/helm/releases/tag/v12.0.0

2021 Feb

13.1.0

Patch notifications, ALS, bulk fixes, enhancements, Testing enhancements

2022 Mar

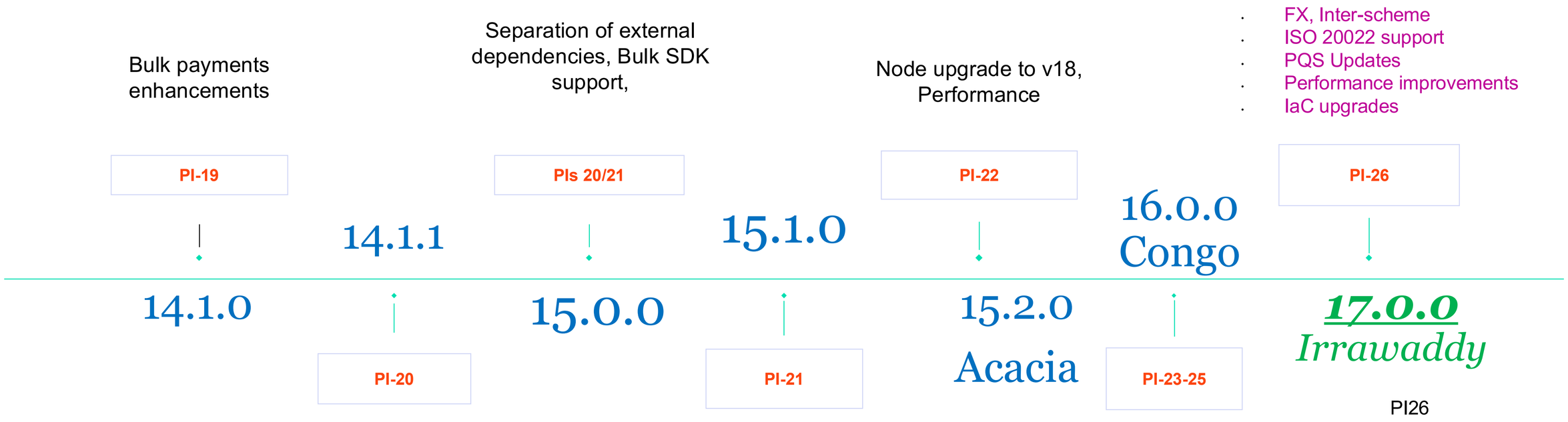
14.0.0

Upgrade to node 16. Maintenance: Security, third-party, TTK, sdk updates

2022 Sep

Mojaloop

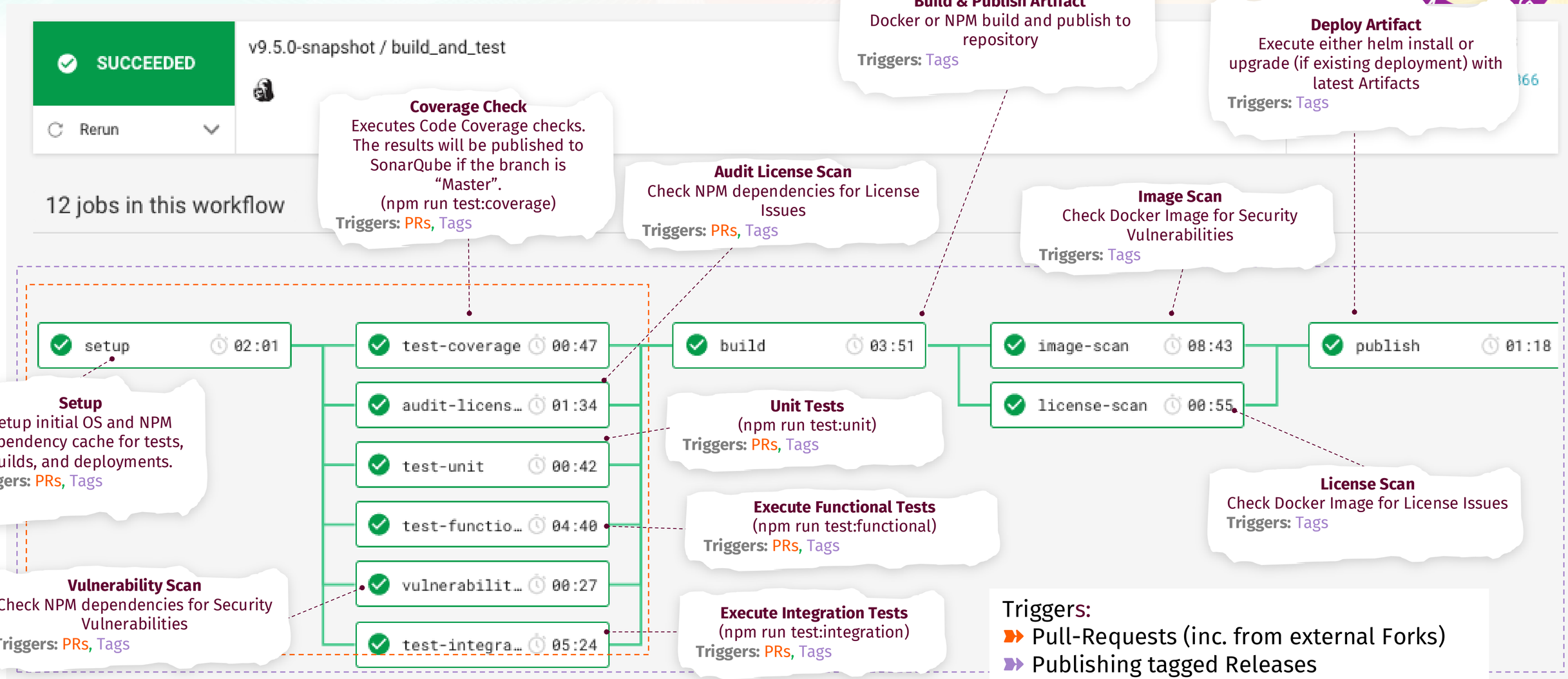
PI-25



** Version numbers guided by SemVer*

Deployment Architecture: CI/CD Pipeline - circleci

Ref https://app.circleci.com/projects/project-dashboard/github/mojaloop



Triggers:

- Pull-Requests (inc. from external Forks)
- Publishing tagged Releases
- Publishing tagged Snapshots